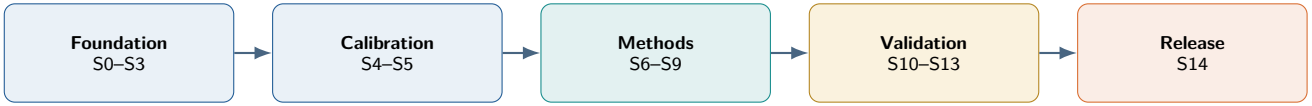
Topological Asteroseismology of Solar-like Stars

Gantt-style research roadmap ▪ 30-month publication program

Persistent structure in light curves, power spectra, and cylindrical échelle geometry

Execution logic



Primary scientific outputs

Output	Scientific nucleus	Target evidence
Paper I	Persistent topology of solar-like oscillation spectra	Synthetic recovery, Rudy benchmark, first locked Kepler confirmation
Paper II	Cylindrical topological échelle geometry	Ridge coherence, $\Delta\nu$, small separations, stability bounds, LEGACY validation
Paper III	Hybrid inference and Kepler–TESS transfer	Conditional information, quality flags, degradation tests, actual TESS data
Software	Reproducible topological-asteroseismology toolkit	Versioned workflows, benchmark notebooks, archived data products

Planning conventions

Project months M1–M30 are the authoritative full-program axis. They may be mapped onto any formal start date without changing dependencies.

DEFENSE-CRITICAL Must finish for the May 2027 thesis core.

CORE Required for the principal scientific claims.

DECISION GATE Advance, repeat a bounded task, or stop/reframe.

EXTENSION High-value work that cannot endanger the thesis core.

Defense-critical outcome

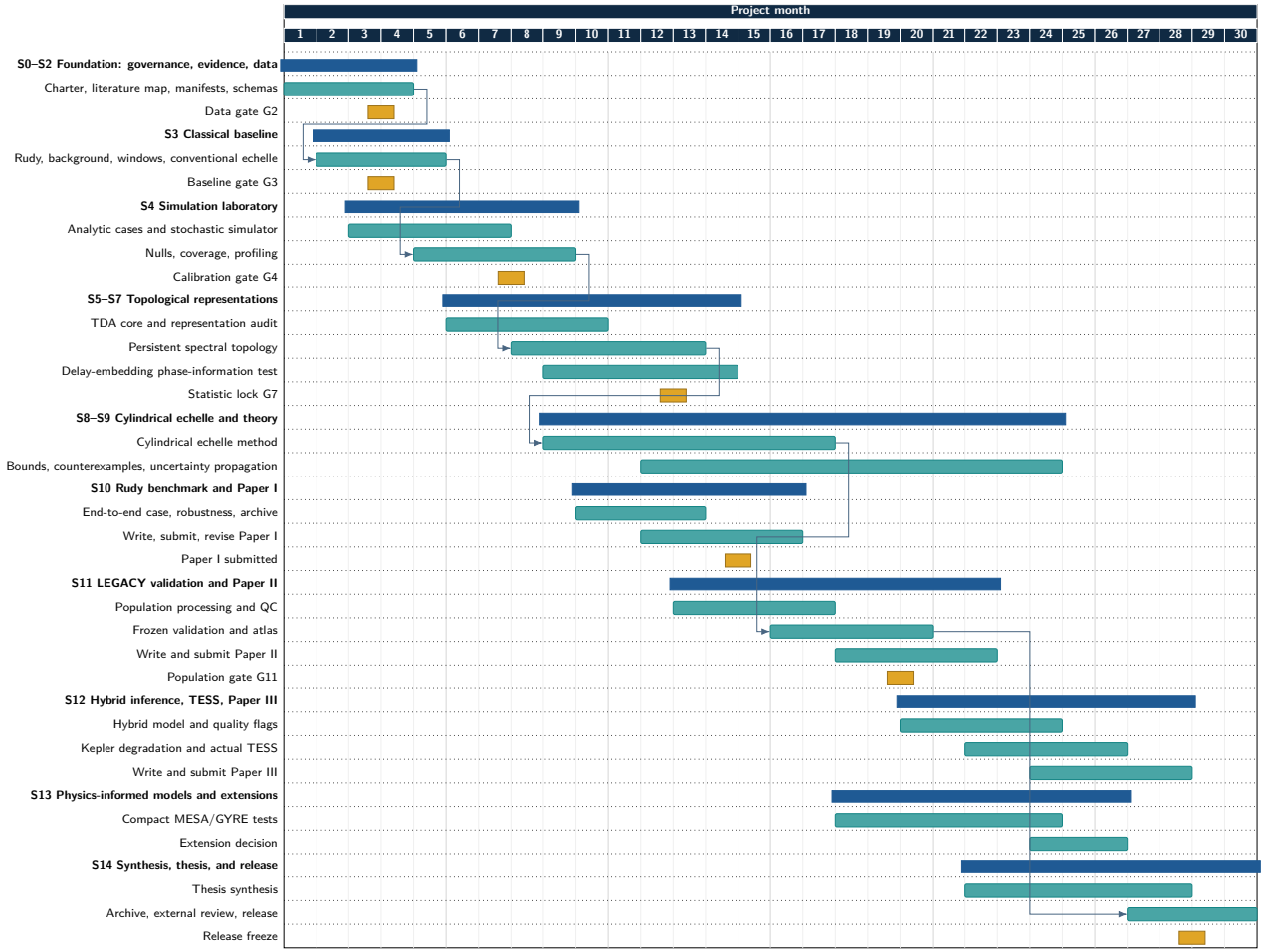
By May 2027: a mathematically specified and null-calibrated framework; a complete KIC 10963065 (Rudy) benchmark; one independently locked Kepler confirmation; thesis-ready evidence, code, and limitations.

Scope rule

After February 2027, no new representation, broad TESS sample, large stellar-model grid, or high-dimensional learner enters the thesis core unless it repairs a correctness defect.

Full 30-month Gantt roadmap

Primary concentration of work; testing and documentation run continuously



FOUNDATION

Establish provenance and a trusted conventional baseline.

METHOD DEVELOPMENT

Calibrate every retained statistic against truth and topology-free controls.

MILESTONE / GATE

Advancement requires prespecified evidence; a rigorous null result is acceptable.

VALIDATION / RELEASE

Lock configurations before population and mission-transfer tests.

Defense-critical path: August 2026–May 2027

Compressed thesis plan; full publication program continues after defense

Month	Calendar	Primary work	Required freeze / evidence	Decision status
M1	Aug 2026	S0 governance; S1 evidence map; S2 Rudy manifests and schemas	Charter, protocol, data identities, benchmark specification	CRITICAL
M2	Sep 2026	S3 Lightkurve/pySYD baseline; background, window, and echelle audits	Classical reference, reproducibility report, Gate G3	CRITICAL
M3	Oct 2026	S4 analytic cases, stochastic simulator, null families, first coverage study	Valid simulator, null design, truth-recovery tests	CRITICAL
M4	Nov 2026	S5 TDA core; S6 spectral persistence; topology-free ablations	Retained statistics, calibrated summaries, failure taxonomy	CRITICAL
M5	Dec 2026	S8 cylindrical echelle prototype; Rudy robustness branches	Topological method v0.9; frozen pilot factors	CRITICAL
M6	Jan 2027	S10 end-to-end Rudy synthesis; compact Kepler development subset	Rudy result package; pipeline QC and candidate claims	CRITICAL
M7	Feb 2027	Locked Kepler confirmation; S9 core proofs and bounds	Confirmation tables; theorem/bound statements; claim ledger	SCOPE FREEZE
M8	Mar 2027	Paper I submission; thesis chapter and figure freeze	Manuscript, evidence matrix, archived configuration	CRITICAL
M9	Apr 2027	Thesis submission; independent reproduction; defense rehearsal	Thesis release candidate; reproduction report	CRITICAL
M10	May 2027	Corrections, archival deposition, seminar, defense buffer	Thesis and software v1.0 defense release	DEFENSE

Minimum success criterion

A defensible thesis does not require TDA to outperform every classical pipeline. It requires a mathematically explicit, statistically calibrated result showing either (i) stable complementary information or quality control, or (ii) a rigorous boundary demonstrating redundancy under specified conditions.

Protected post-defense backlog

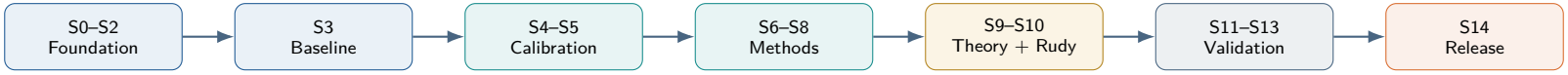
Full 66-star LEGACY atlas; broad Kepler–TESS transfer; extensive MESA/GYRE grids; acoustic glitches; rotation and inclination; large hybrid models; complete Papers II and III. These remain valuable, but cannot delay thesis submission.

Important: Calendar mapping assumes launch in August 2026. If the start date changes, preserve stage order and gates; remap dates rather than silently compressing calibration or locked validation.

Stage register and dependencies

Each stage advances only after its deliverables and gate evidence are accepted

Stage	Purpose	Main months	Key deliverables	Advance criterion / dependency
S0	Governance and reproducibility foundation	M1–M2	Charter, repository, environments, conventions, claim ledger	Scope, ownership, versions, and review rules approved
S1	Literature evidence map and benchmark specification	M1–M3	Evidence table, gap statement, benchmark and hypothesis registry	Every central claim linked to evidence and a falsifiable test
S2	Data acquisition, provenance, canonical data model	M1–M4	Immutable manifests, schemas, QC, Rudy/LEGACY/TESS identities	Checksums and provenance complete; no unresolved unit/time ambiguity
S3	Classical asteroseismic baseline and observation model	M2–M5	Lightkurve/pySYD results, PSD/background/window/echelle audit	Rudy values reproducible and discrepancies explained
S4	Analytic cases and stochastic simulation laboratory	M3–M9	Damped-mode simulator, gaps/noise/window factors, null families	Truth recovery, Type-I error, coverage, and profiling acceptable
S5	TDA core and representation audit	M6–M10	Validated persistence engine, summaries, uncertainty and ablations	Agreement on analytic cases; no representation artifact dominates
S6	Persistent topology of the power spectrum	M8–M13	Peak hierarchy, topological $\nu_{\max}/\Delta\nu$ candidates, null calibration	Competitive recovery or demonstrable robustness / reliability value
S7	Delay topology and phase-information test	M9–M14	Embeddings, persistence, PSD-preserving surrogate comparison	Retain only with phase-sensitive or lifetime/coherence information
S8	Cylindrical topological echelle diagram	M9–M17	Periodic-boundary filtration, ridge graph, coherence score	Correct boundary behavior and stable ridge recovery on truth/Rudy
S9	Mathematical relations, bounds, counterexamples	M12–M24	Stability bounds, ridge-drift results, uncertainty propagation	Assumptions explicit; numerical stress tests support theorem scope
S10	Rudy end-to-end benchmark and Paper I freeze	M10–M16	Locked case study, robustness matrix, archive, Paper I	All claims trace to prespecified tests; independent rerun succeeds
S11	Kepler LEGACY population validation	M13–M22	Star-level splits, QC atlas, locked metrics, Paper II	Generalization shown or boundary characterized without leakage
S12	Hybrid inference, QC, Kepler–TESS transfer	M20–M28	Conditional tests, quality flags, degradation and TESS results	Gain exceeds uncertainty/practical margin or useful QC certified
S13	Physics-informed stellar models and extensions	M18–M26	Compact MESA/GYRE tests, ranked extension decisions	Proceed only when core is stable and extension has discriminating power
S14	Synthesis, publications, software, thesis	M22–M30	Papers, thesis, software release, archive, documentation	Reproduction, licensing, evidence matrix, and release checklist pass



Milestones, gates, and control rules

Evidence-driven decisions protect both rigor and schedule

Month	Review	Required evidence	Decision
M1	Launch review	Charter, definitions, repository, access, risks	Approve scope
M3	Classical gate	Rudy reproduction, QC, spectral windows, reference values	Authorize topology
M5	Simulator review	Truth recovery, analytic tests, null validity	Freeze pilot factors
M7	Calibration gate	Type-I error, interval coverage, compute profile	Scale or simplify
M9	Representation review	Localization, redundancy, topology-free ablations	Retain or drop
M12	Statistic lock	Estimators, thresholds, blinded Rudy/Kepler configuration	Authorize confirmation
M14	Paper I gate	Rudy archive, independent rerun, complete evidence matrix	Submit Paper I
M19	Population gate	Locked LEGACY results, subgroup and failure analysis	Generalize or bound
M24	Hybrid/extension gate	Conditional information, practical gain, model evidence	Retain or stop
M28	Release freeze	Papers, thesis draft, software, archive, documentation	External review
M30	Program close	Reproduction package, limitations, backlog handoff	Release / defend

Gate outcomes

ADVANCE All mandatory criteria pass; freeze inputs and proceed.

BOUNDED REPEAT A named, correctable defect exists; repeat only specified tasks with an unchanged holdout and deadline.

STOP / REFRAME The method is redundant, unstable, uncalibrated, or infeasible; record the boundary and move effort to the strongest surviving path.

Non-negotiable controls

- Split and validate by star, never by segments of the same star.
- Freeze estimators and thresholds before locked confirmation.
- Compare against classical and topology-free baselines.
- Use background-only, randomized-comb, phase-randomized, and IAAFT nulls.
- Report failures, non-detections, aliases, and selection effects.
- Treat rigorous negative results as valid scientific outputs.

Operational principle. The roadmap optimizes for defensible scientific claims, not for completion of every proposed technique. A stage is complete only when its deliverables are reproducible and its advancement evidence is recorded; elapsed time alone never closes a stage.